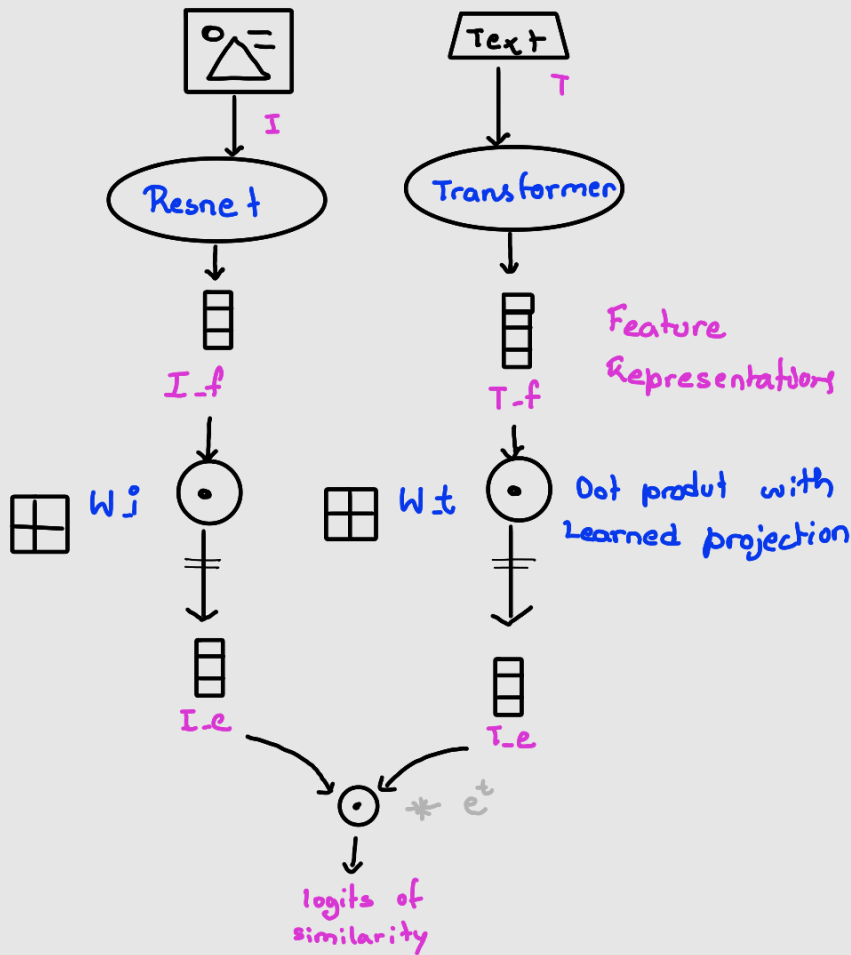


CLIP - Contrastive Language - Image Pretraining



```
# image_encoder - ResNet or Vision Transformer
# text_encoder - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l] - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T) #[n, d_t]

# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss = (loss_i + loss_t)/2
```

Figure 3. Numpy-like pseudocode for the core of an implementation of CLIP.

This is **computed** for all image text pairs

↑	↓	↓	↓
↓	↑	↓	↓
↓	↓	↑	↓
↓	↓	↓	↑

$n \times n$

logit

→ cross entropy (logit, $\begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$)

Image encoders : 5 ResNet 3 ViT models

Text encoders : Transformer (BPE, 49k vocab)

DINO - Self distillation with NO labels

Student Network (g_{θ_s})
GD

Teacher Network (g_{θ_t})
EMA

Multi crop strategy training.

- Teacher gives global view embedding which student tries to estimate using local views

Architecture :

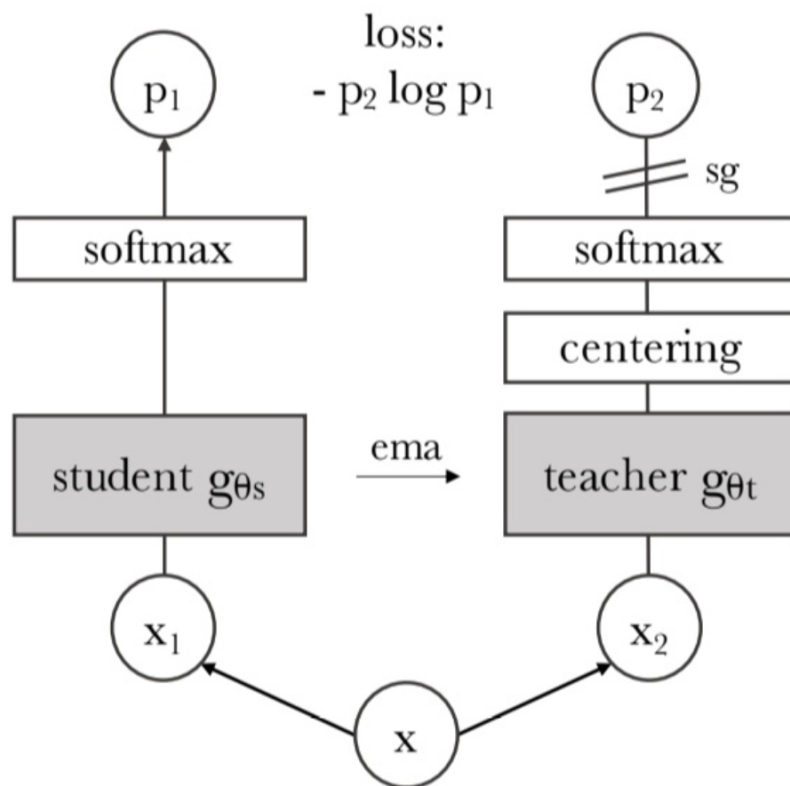
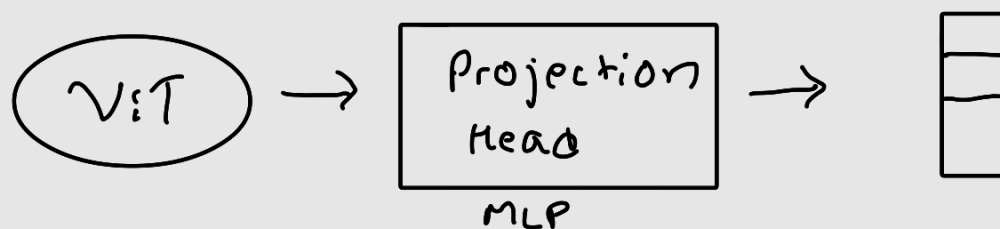
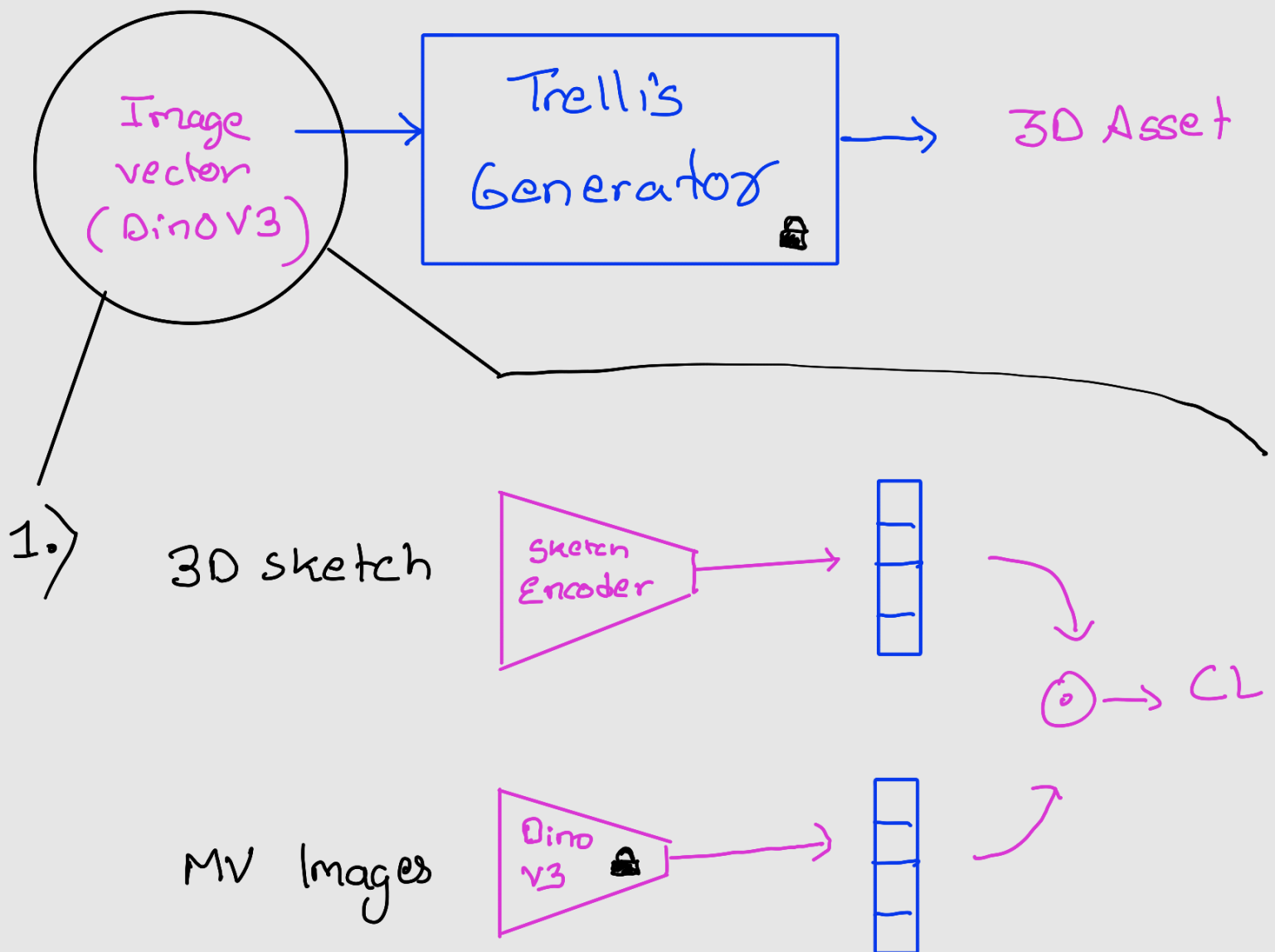


Figure 2: **Self-distillation with no labels.** We illustrate DINO in the case of one single pair of views (x_1, x_2) for simplicity. The model passes two different random transformations of an input image to the student and teacher networks. Both networks have the same architecture but different parameters. The output of the

the same architecture but different parameters. The output of the teacher network is centered with a mean computed over the batch. Each networks outputs a K dimensional feature that is normalized with a temperature softmax over the feature dimension. Their similarity is then measured with a cross-entropy loss. We apply a stop-gradient (sg) operator on the teacher to propagate gradients only through the student. The teacher parameters are updated with an exponential moving average (ema) of the student parameters.

How to use them?

Data: { 3D asset , Images , Sketches }



2.)

3D Sketch → Render Images → Image 2 Image
Diffusion
Model

